SONYLIV CLICK-A-THON · CLICKHOUSE

The Phoenix

Concurrent viewers, counted correctly. Peak and average at minute, hour and day grain, across nine filter dimensions, from a delta table that never rescans raw history.

905,558
GRADED CORPUS EVENTS

7,000,000
UNSEEN DAY, SAME QUERIES

0
DIFFS AGAINST BRUTE-FORCE ORACLE

THE PROBLEM

Interval overlap is not concurrency

A session can be open while the app is backgrounded, the player is paused, or the heartbeat has simply stopped arriving. Counting that time as watching overstates the audience, and every decision made on the dashboard inherits the error.

A **31 percent** overstatement, and 1,592 minutes where the naive method invents an audience that was never there. That gap is the entire problem, removed and measured.

THE MODEL

Model the active interval, not the session

Every event moves a session between three states: foreground, background, and gone. State holds for a 90 second tolerance and no longer, so a session that stops emitting stops counting whether or not it ever formally ended.

Those active intervals normalise to per-session minute runs, then to `+1/-1` boundary deltas. Concurrency at any minute is a running sum, so the serving layer reads boundaries rather than history.

- **Peak is computed, never stored.** A platform slice and a platform-plus-country slice peak at different minutes inside the same range, so a pre-rolled peak per dimension is wrong by construction.
- **Both averages are reported.** 88.06 over all 1,440 minutes, 200.00 over the 634 minutes with an audience. Averaging sparse rows instead returns 246.5 against a true 88.2.
- **Open sessions retract.** A session still open when the question is asked is counted, then corrected as its real ending arrives.

ARCHITECTURE

Raw history is read once, on the way in

CSV		event_state		concurrency_deltas		serving		console
905,558 events, landing table with ENGINE = Null so nothing is stored twice.	→	Three-bucket state machine, 90 second gap tolerance, foreground only.	→	SummingMergeTree of +1/-1 boundaries keyed by minute and nine dimensions. 439 KiB for the unseen day.	→	Seeded cumulative sum. Peak, both averages, p95, at any grain.	→	Prints the query, the table it read, and the rows read, next to every answer.

Nothing in the serving path touches `raw_events`. The gate that proves it reads ClickHouse's own `query_log` and fails the build if a served query ever does.

`./scripts/check_query_sources.sh`

RESULTS · CORRECTNESS

The numbers are not asserted

CLAIM	NUMBER	REPRODUCE
Serving layer versus brute-force oracle	**3,663 minutes, 0 diffs**	parity.sh
Open sessions absorbed incrementally	**5,316 minutes, 0 diffs**	test_open_sessions.sh
Full rebuild run twice, derived tables diffed	**0 diff lines, 5 of 5 tables**	prove_idempotence.sh
Frozen slice stable under concurrent writes	**34 metrics, 0 differing lines**	frozen_gate.sh
Peak concurrent sessions, graded corpus	**2,828 at 10:56**	ground_state.sh
Average over all 1,440 minutes	**88.06**	bench.sh
Average over the 634 minutes with an audience	**200.00**	bench.sh

Every figure resolves through `evidence/LEDGER.tsv` to the command that produced it and the stamped artifact it wrote. `check_docs.sh` fails the build when a claim and its evidence disagree.

Fast because it reads less, not because it is bigger

12 ms

WORST-SHAPE QUERY,
30,662 ROWS AGAINST A BUDGET OF 80,712

2 of 4

GRANULES TOUCHED BY A
PLATFORM FILTER, 16,384 ROWS

70 s

CSV TO VERIFIED SERVING LAYER,
FULL REBUILD FROM COLD

The sorting key is ordered by pruning power, so the dimensions a judge actually filters on come first. The honest caveat, stated in the README rather than hidden: a ninth-position dimension such as `video_resolution` does not prune, and reads 354,185 rows against 354,305 unfiltered.

Released after the pipeline was built. Same queries, untouched.

7,000,000	41 s	22,416	914.65
EVENTS FOR 2026-07-31, 7.7X THE GRADED CORPUS	TO DERIVE, ALL INVARIANTS PASS	PEAK CONCURRENT SESSIONS AT 11:16	DAILY AVERAGE OVER 1,440 MINUTES

Both datasets sit behind a switch in the console, so the generalisation can be seen rather than claimed: the original corpus every number was measured against, and the unseen day, side by side, without a restart.

UPDATE-FRIENDLY

Nine dimensions cost 98 KiB and change no answer

The same 7,000,000 row day, derived twice: five dimensions against nine. The widened key was accepted only because the comparison came back identical, line for line.

METRIC	FIVE DIMENSIONS	NINE DIMENSIONS
Peak concurrent	22,416 at 11:16	22,416 at 11:16
Average, all minutes	914.65	914.65
Average, active minutes	1,667.21	1,667.21
Asserted runs	118,498	118,498
Derive time	41 s	41 s
Delta table on disk	341 KiB	439 KiB

Key cardinality rises 1.6x, realised rows only 11 percent, because SummingMergeTree collapses what the wider key splits.

An evidence ledger, not a claims list

- **Every script writes a stamped TSV.** Run timestamp, git SHA, host, row count, watermark. A claim with no artifact behind it fails `check_docs.sh`.
- **Gates read ClickHouse's own query_log** to prove which tables a served query touched, rather than trusting the SQL to say so.
- **Corrections are kept, not erased.** A positional INSERT silently shifted a column while all three derive invariants passed. That is written down, with the fix: explicit column lists everywhere.
- **Two consoles on one port.** The graded concurrency console, and a live insight console where the curve builds in near real time from a running producer.
- **A live agent, not a canned demo.** The Ask tab is backed by a LibreChat agent with a real ClickHouse MCP tool, so an open-ended question is answered by querying, not by a script.
- **Deployed behind one published port.** A check fails the deploy if anything else is exposed.

LIVE NOW

See it answer

the-phoenix.cricheroes.io serves the graded console at `/` and the live insight console at `/v2`. Both print the query, the table, and the rows read beside every answer, so nothing on screen has to be taken on trust.

31%	0	12 ms	41 s
OVERSTATEMENT REMOVED AND MEASURED	DIFFS AGAINST THE ORACLE, BATCH AND INCREMENTAL	WORST-SHAPE QUERY, NINE FILTER DIMENSIONS	TO ABSORB AN UNSEEN SEVEN MILLION EVENT DAY